Marissa Kivi

LinkedIn

High-impact data scientist specializing in end-to-end data-model pipelines that support agtech and carbon MRV products. Combines deep domain expertise in agronomy and ecosystem modeling with a strong commitment to data integrity and scientific rigor. Skilled at rescuing complex data initiatives, establishing end-to-end data governance, and delivering scalable, production-grade solutions.

Skills

Tools: Python, SQL (PostgreSQL), R, Bash, Git, Docker, AWS (RDS, S3), Kubernetes, Leaflet, Claude Code

Data Engineering & Pipelines: Geospatial Data Pipeline Development, Schema Design, Data Validation & QA/QC, Remote Sensing and Satellite Imagery Analysis, ETL/ELT Workflows

Machine Learning & Analytics: Supervised Regression & Classification, Tree-Based Models, Time Series Analysis, Feature Engineering, Model Evaluation & Monitoring, Statistical Analysis, Uncertainty Quantification, Data Assimilation

Domain & Process Modeling: Soil Carbon MRV, Ecosystem Models (APSIM, DayCent, LINKAGES)

Data Governance & Strategy: Data Auditing, Codebase Refactoring & Optimization, Process Mapping, Product Data Strategy, Technical Documentation

Work Experience

Modeling Scientist
Arva

Houston, TX (Remote)
Aug 2025 – Current

- Led a data governance audit of a critical Scope 3 contract fulfillment pipeline across 5+ cross-functional teams, documenting data flows, QA/QC validation rules, and decision points; presented risk assessment and final recommendation to executive leadership.
- Rescued a failing project within two months by refactoring the core codebase, reducing complexity from ~13K to ~6K lines of code while improving maintainability, readability, and performance.
- Designed and implemented a data validation pipeline with internal tooling appropriate for non-technical users, enabling early and systematic detection of data issues and improving downstream simulation reliability.
- Containerized a third-party model executable to standardize execution across the modeling pipeline, ensuring that our team met critical delivery deadlines.
- Re-designed database schema to be generalizable and scalable, supporting future projects and reducing data ambiguities and rework.
- Helped establish lightweight project management structures across projects to improve communication, prioritize work, and track progress more effectively.
- Authored and shared internal documentation covering data schema and simulation setup logic, improving cross-team alignment and data traceability.
- Mentored junior-level analysts in building programming skills and data management instincts.

Senior Data Scientist
EarthOptics Inc.

Minneapolis, MN (Remote)
2023-2025

- Conducted and presented analyses of experimental datasets to support product discovery, validation, and go/no-go decisions.
- Designed validation and preprocessing pipelines for high-dimensional sensor data to improve signal clarity and reduce time needed to identify malfunctioning equipment.

- Refactored the automated ML pipeline in production to reduce training time, improve flexibility for rapid iteration, and increase visibility into model performance over time.
- Led cross-functional definition and prioritization of product data requirements, improving product quality and strengthening internal data literacy.
- Led development of internal software tools to improve reproducibility, automation, and accuracy in internal QA/QC processes.

Machine Learning Scientist
Sentera

St. Paul, MN (Hybrid)
2022-2023

- Built ML models to support customer-defined agronomic objectives through prediction and pattern analysis.
- Designed and maintained a PostgreSQL database integrating agronomic, satellite, and climate data to improve data accessibility and delivery speed.
- Engineered a reproducible ML pipeline in Python using Metaflow and AWS compute services to improve development efficiency.
- Developed data processing pipelines for weather data, satellite and drone imagery, and agronomic field data.
- Created data engineering strategies to augment limited customer data inventories and improve data utility.
- Supported project scoping and led customer-facing discussions and presentations of results.

Graduate Research Assistant
Department of Crop Sciences
University of Illinois

Urbana, IL
2020-2022

- Iteratively developed an agricultural forecasting framework combining a process-based crop model (APSIM) with observed field data to sequentially optimize model performance and estimate uncertainties.
- Compiled and cleaned diverse agricultural datasets across spatial and temporal scales into a unified analysis-ready dataset.
- Designed and evaluated a statistical model to estimate field-scale planting dates from features derived from satellite-based NDVI time series.

Research Assistant
Department of Biology
University of Notre Dame

Notre Dame, IN
2019-2020

- Modeled forest community dynamics and aboveground biomass using a process-based ecosystem model (LINKAGES) with sequential data assimilation.
- Analyzed simulation outputs using tree-based methods, multivariate regression, and high-dimensional visualization to identify drivers of forest growth.
- Led version control and documentation efforts for data processing and assimilation workflows.

EDUCATION

Master of Science in Crop Sciences, *University of Illinois at Urbana-Champaign* May 2022
Research Assistantship | Concentration in Digital Agriculture | GPA: 4.0/4.0
Kraft-Frerichs Graduate Fellow (2021-2022)

Master of Science in Applied & Computational Mathematics and Statistics, *University of Notre Dame* Jan 2020
Concentration in Predictive Analytics | GPA: 4.0/4.0
Outstanding Applied & Computational Mathematics and Statistics Major Award Recipient

Bachelor of Science in Applied & Computational Mathematics and Statistics, *University of Notre Dame* May 2019
Minor in Sustainability | Summa Cum Laude | Valedictorian Finalist | GPA: 4.0/4.0